Where to Build Next

Not every connectivity gap needs a tower. This siting tool sorts every square kilometre of Nigeria by what it needs, then picks the clinics and schools that should host new sites.

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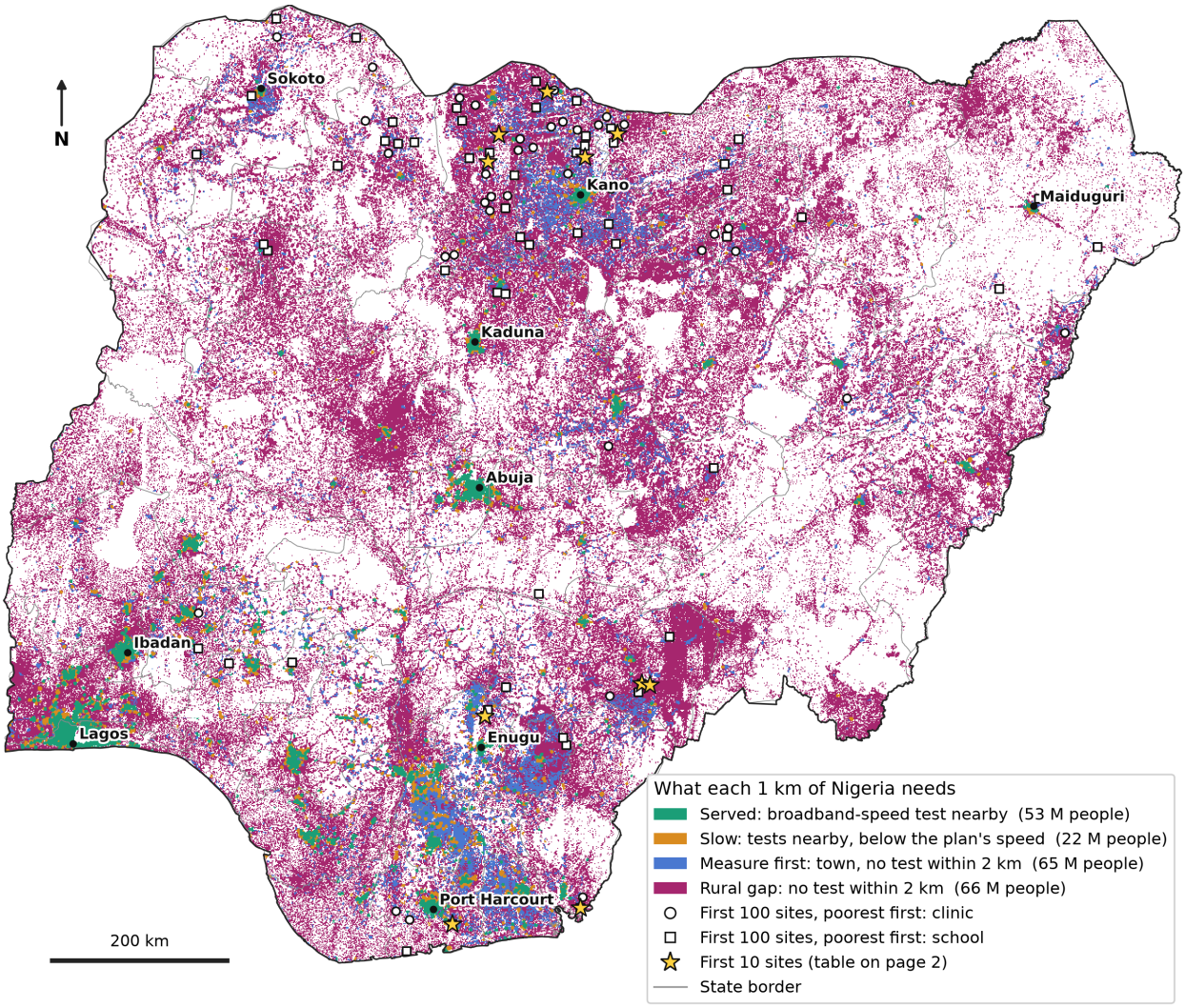
Interactive tool: https://dexsholex.github.io/where-to-build-next/

53.0 M
Served, 26% of people

22.1 M
Slow: upgrade, 11% of people

64.6 M
Unmeasured town, 31% of people

66.0 M
Rural, no tests, 32% of people



Speedtest by Ookla, mobile, Q4 2024; Meta population 2020 and Relative Wealth Index; GHSL settlement classes 2020; GRID3 health and education facilities; OCHA boundaries. Speed benchmark: National Broadband Plan 2020-2025 (25 Mbps urban, 10 Mbps rural). Albers equal-area. Analysis: Oluwasola Oyinloye.

Figure 1. What each 1 km of Nigeria needs, from 42,838 Ookla mobile speed tests (Q4 2024) against the National Broadband Plan's speeds, with the first 100 sites (clinics and schools), chosen poorest first to close rural gaps.

The problem

Nigeria's broadband plan set a 2025 target of 25 Mbps in urban areas and 10 Mbps in rural areas, with 4G for 90% of people (FMoCDE 2020). In the Ookla data, only 2.6% of rural residents live within 2 km of a test at the plan's speed, against 37.6% in towns. Where people do test, the average download is 52.2 Mbps in towns and 22.9 Mbps in rural areas (weighted by tests; latency 56 and 54 ms), but only 628 of the 9,234 tiles are rural. The question this tool answers: which communities are poorly connected relative to their population, settlement and need, what does each place need, and where should the next sites go?

Steps

1. **Identify the data.** DSN's file states neither type nor period. Matching tiles to Ookla's public release on AWS identified it exactly: mobile, Q4 2024 (all 9,234 tiles identical).
2. **Build a 1 km grid.** 205.6 million people from Meta's 2020 population map, each cell tagged urban or rural (GHSL), with a wealth score (Meta RWI) and its LGA (OCHA). 256 pixels where Meta's layer piles a census count onto a few buildings (0.55 million people) are excluded as unplaceable.
3. **Sort every cell by need.** Served: a test within 2 km meets the plan's speed for its setting. Slow: tests nearby, all too slow. Unmeasured town: urban, no test within 2 km. Rural gap: rural, no test within 2 km.
4. **Choose the sites.** For rural gaps only, a maximal covering search (Church and ReVelle 1974) picks hosts from 54,732 registered clinics and 204,355 schools (GRID3; public and private, all levels), one per km², that reach the most people within 5 km (the low end of ITU's 5 to 15 km range for mobile cells); greedy selection reaches at least 63% of the optimum (Nemhauser et al. 1978). It runs twice: reach the most people, and poorest first (the poorer 40% by wealth score). JAMB centres have no official list with locations.

External datasets integrated with the Ookla tiles

Meta population 2020	people per ~31 m pixel, summed to 1 km	Facebook and CIESIN 2021
Meta Relative Wealth Index	need: poorer 40% of people	Chi et al. 2022
GHSL degree of urbanisation 2020	urban or rural, sets the speed threshold	Schiavina et al. 2023
GRID3 health facilities v3.0, v2.0	54,732 candidate clinic hosts	CIESIN 2026, 2024
GRID3 education facilities 2020	204,355 candidate school hosts	GRID3 2020
OCHA boundaries; Broadband Plan	LGAs; speed and coverage benchmarks	OCHA FISS 2026; FMoCDE 2020
OpenCelliD cell positions 2024	check: is any mobile cell recorded near a gap?	OpenCelliD 2024

Insights and recommendations

1 **Rural Nigeria is nearly invisible in the speed data, and its gap is a poverty gap.** Towns log 2.99 tests per 10,000 people, rural areas 0.31. 66.0 million rural people have no test within 2 km, and 80% of them are in the poorer 40%. The LGAs holding most of them (table below) are thinly settled: 126 to 210 poorer people per km² of gap, against 482 under the 100 chosen sites, so in 4 of the 5 even the best site reaches fewer people than the 100th site chosen.

Action. Universal Service Provision Fund: give these LGAs a budget line of their own, as each needs several sites.

2 **Most gaps need upgrades or measurement, not new towers.** 22.1 million people live near tests that are too slow (only 38.9% of town tiles reach 25 Mbps), and 64.6 million live in towns with no test at all; 231 of 774 LGAs (48.9 million people) had none.

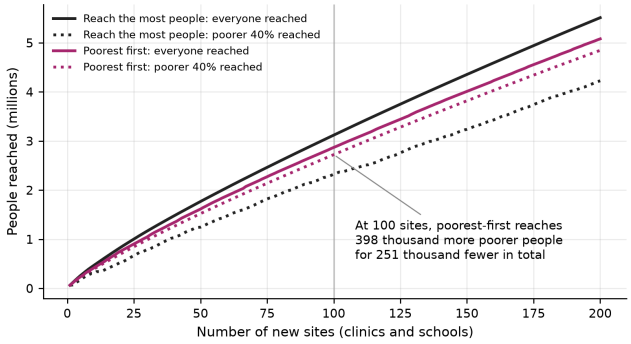
Action. Operators: add capacity where tests are slow. Nigerian Communications Commission: measure untested towns first.

3 **Clinics and schools can share the new sites, and choosing the poorest first costs little reach.** 21,269 clinics and 76,665 schools sit in rural gaps. 100 sites at either reach 3.12 million people, against 3.04 million with clinics alone. Chosen poorest first, they reach 2.87 million: 398 thousand more people in the poorer 40% (17% more) for 251 thousand fewer overall; 60 of them are schools, and 86 of the 100 squares hold two or more possible hosts.

Action. NPHCDA, state primary health care and universal basic education boards, and the Fund: host new sites at these clinics and schools, chosen poorest first, towards the plan's target of connecting 80% of primary health centres.

Communities to prioritise (LGAs, people in thousands)

Rural, poorer 40%, no broadband evidence		Towns to measure first		Slow networks to upgrade	
1. Ningi, Bauchi	385	1. Okene, Kogi	524	1. Oluyole, Oyo	315
2. Fune, Yobe	321	2. Idemili North, Anambra	428	2. Ifo, Ogun	304
3. Wasagu/Danko, Kebbi	315	3. Ihiala, Anambra	391	3. Aguata, Anambra	284
4. Ganjuwa, Bauchi	315	4. Andoni, Rivers	351	4. Idemili South, Anambra	265
5. Gamawa, Bauchi	309	5. Gwoza, Borno	336	5. Ado Ekiti, Ekiti	263



How sure are we?

With 100 sites, reach is 1.7 to 8.0 million for a site reach of 3 to 10 km; the rural gap is 59.4 to 68.0 million across a 1 to 5 km serving distance and a 3-test minimum. Two checks suggest the gaps are not just untested: 95.9% of people in them had no test within 2 km in any quarter of 2024, and in OpenCelliD's crowdsourced cell map (mostly 2G records) 74.5% have no mobile cell of any kind within 5 km, against 3.9% in served areas. Figure 2: people reached as sites are added.

In the tool

Change the number of sites, the priority and the hosts; read the full ranked list or copy it as CSV; rank LGAs by each need; and turn on 2023 satellite imagery to zoom in on any site and check the ground before building.

The first 10 sites, chosen poorest first (full list and both runs in the tool)

#	Host	LGA, state	People reached	In poorer 40%
1	LGEA Primary School Gbandepev (school)	Ushongo, Benue	63,447	63,447
2	LGEA Primary School Ionguga (school)	Ushongo, Benue	44,176	44,176
3	Community Secondary School Bolo (school)	Ogu/Bolo, Rivers	48,801	39,736
4	Government Migrants Secondary School (school)	Akpabuyo, Cross River	42,899	37,880
5	Government Day Secondary School Tsabu (school)	Mai'adua, Katsina	37,611	37,611
6	Dokau Mai Rimaye Health Centre (clinic)	Charanchi, Katsina	36,734	36,734
7	Stella Maris College Ugwu Eke Breme Umuabor (school)	Nsukka, Enugu	44,337	36,423
8	Bakane Health Centre (clinic)	Matazu, Katsina	33,054	32,974
9	Shiddar Health Post (clinic)	Dambatta, Kano	32,943	32,943
10	Dankano Primary School (school)	Babura, Jigawa	33,681	32,639

Limitations

No test means no evidence, not no signal: gaps are inferred from where people test, not from operator coverage maps or mast locations, and tests are crowdsourced and sparse. The needs use Q4 2024 alone (Q1 to Q3 only test that gaps persist; newer quarters exist). A 5 km site reach and 2 km serving distance are assumptions, tested above. Wealth estimates date from 2021. A registered facility is not always a working one, and towers need power, backhaul and security that this tool does not model.

References

Chi, G. et al. (2022) PNAS 119(3) e2113658119. Church, R. and ReVelle, C. (1974) Papers of the Regional Science Association 32, 101-118. GRID3 (2020) NGA Education Facilities, HDX. CIESIN (2024, 2026) GRID3 NGA Health Facilities v2.0, v3.0, doi:10.7916/kv1n-0743, 10.7916/s6cq-1q98. Facebook and CIESIN (2021) High Resolution Settlement Layer. FMoCDE (2020) Nigerian National Broadband Plan 2020-2025. Nemhauser, G., Wolsey, L. and Fisher, M. (1978) Mathematical Programming 14, 265-294. OCHA FISS (2026) Nigeria subnational boundaries. Schiavina, M., Melchiorri, M. and Pesaresi, M. (2023) GHS-SMOD R2023A, JRC, doi:10.2905/A0DF7A6F-49DE-46EA-9BDE-563437A6E2BA. Speedtest® by Ookla® Global Fixed and Mobile Network Performance Maps; based on Oluwasola Oyinloye's analysis of these maps for Q4 2024; Ookla trademarks used under license and reprinted with permission (CC BY-NC-SA 4.0; maps here shared under the same licence). OpenCelliD (2024) cell database, CC BY-SA 4.0, opencellid.org, via Source Cooperative. ITU (2022) Guide for procuring last-mile connectivity data networks, Table 4.